



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

Lecture 19: Gravitation

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Recap: Gravitation

- Newton's Law of Gravity: $F = G \frac{Mm}{r^2} = m \frac{v^2}{R}$

- Circular orbits:

- $v = \sqrt{\frac{GM}{R}}$

- $T = \frac{2\pi R}{v} = \frac{2\pi R^{3/2}}{\sqrt{GM}}$

$$\omega = \frac{v}{R} = \sqrt{\frac{GM}{R^3}}$$

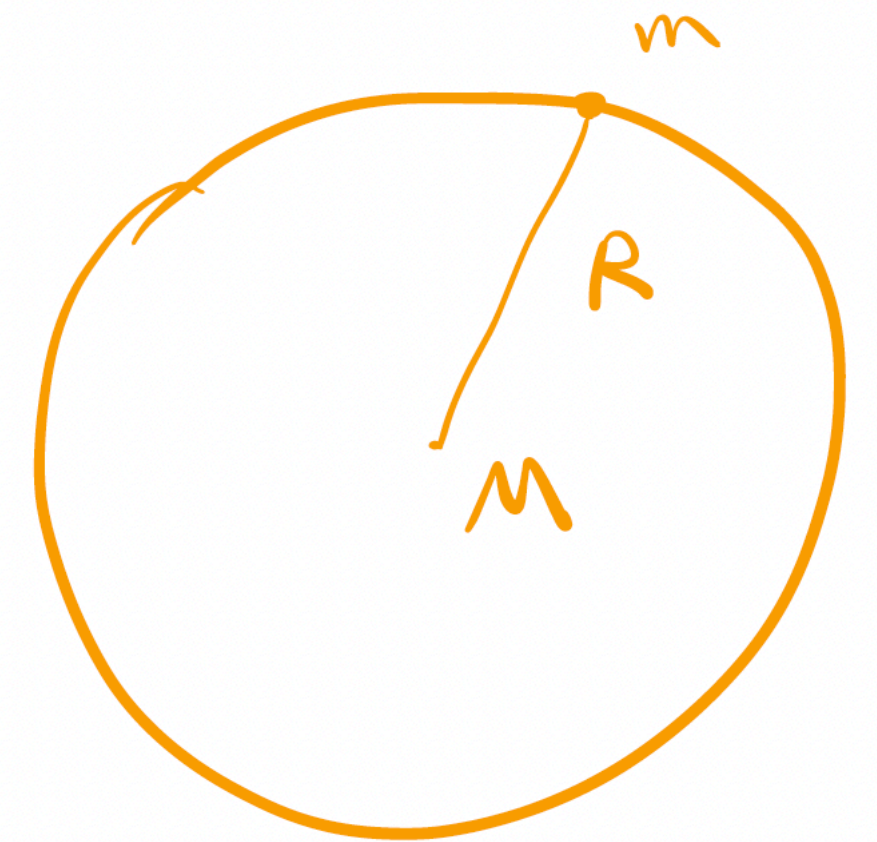
- Gravitational potential energy: $U_g = -G \frac{Mm}{r}$

- Escape velocity $V_e = \sqrt{\frac{2GM}{R}}$

$$\begin{aligned} \rightarrow E > 0 \\ \hookrightarrow V_0 > \sqrt{\frac{GM}{R}} \end{aligned}$$

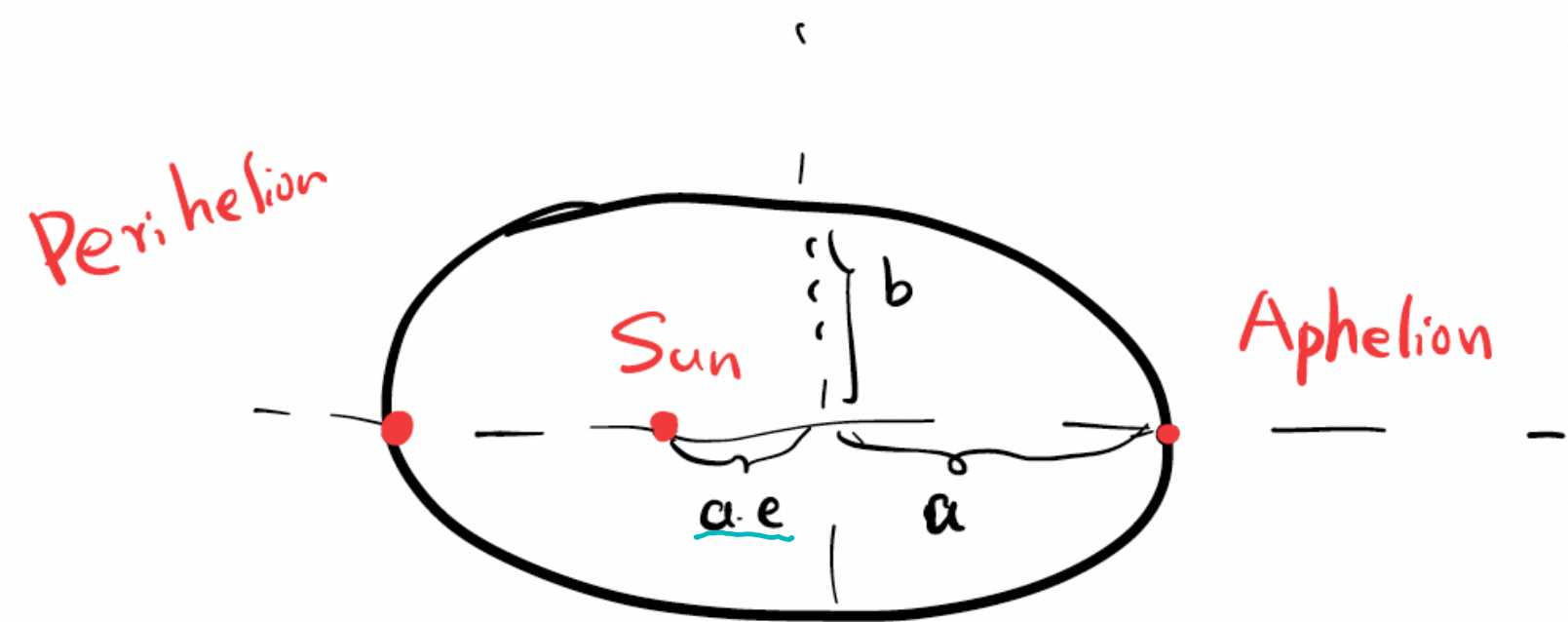
$$E = K + U = \frac{1}{2} m \frac{GM}{R} - G \frac{Mm}{R} = -\frac{1}{2} \frac{GMm}{R} < 0$$

bound state



Kepler's Laws

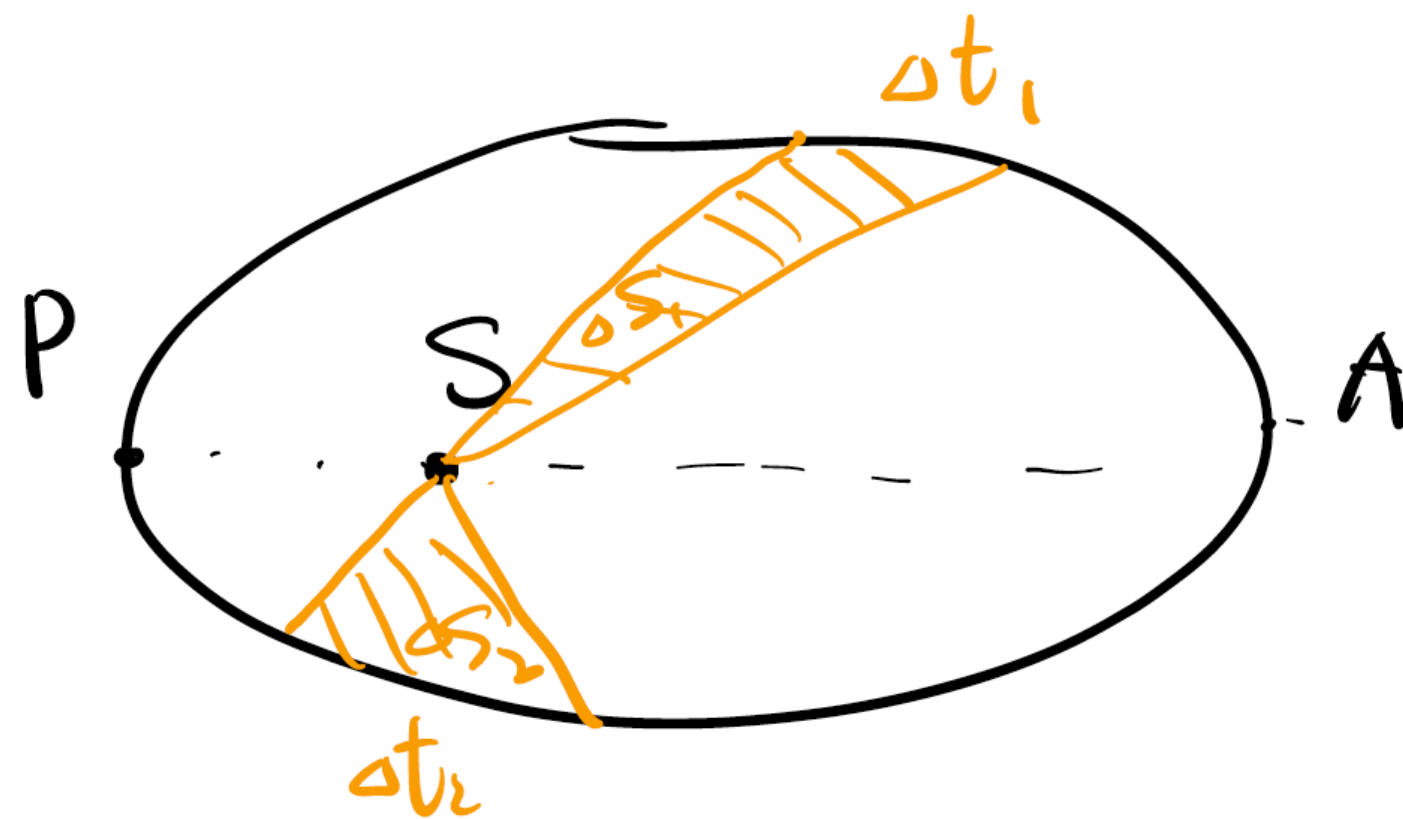
- First law: Orbits are ellipses with the Sun at the focus ^(non-moved)



e : eccentricity
 a : semi-major axis
 b : semi-minor axis

Kepler's Laws

- Second Law: equal time, equal area

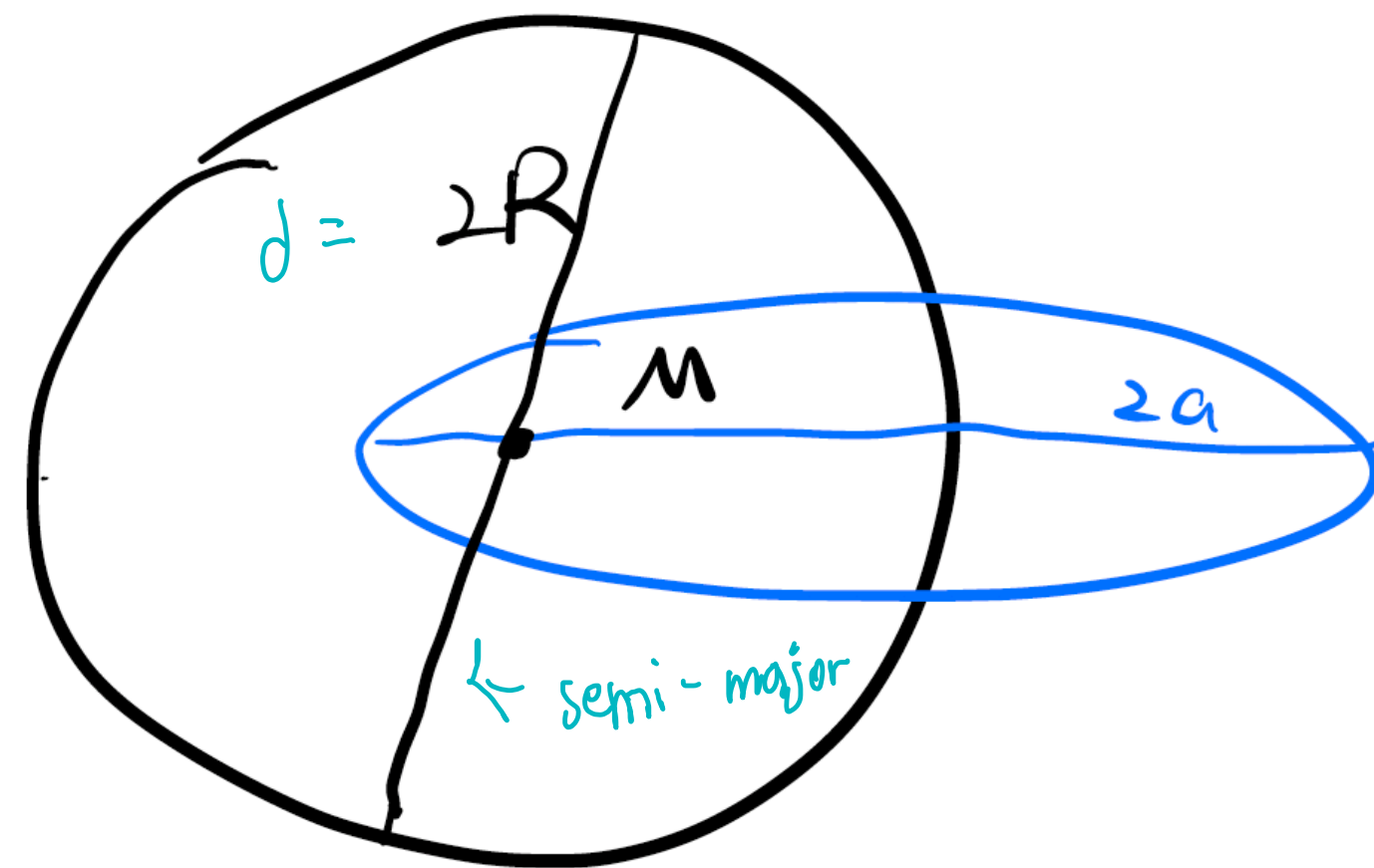


$$\Delta t_1 = \Delta t_2$$
$$\updownarrow$$
$$\Delta S_1 = \Delta S_2$$

Kepler's Laws

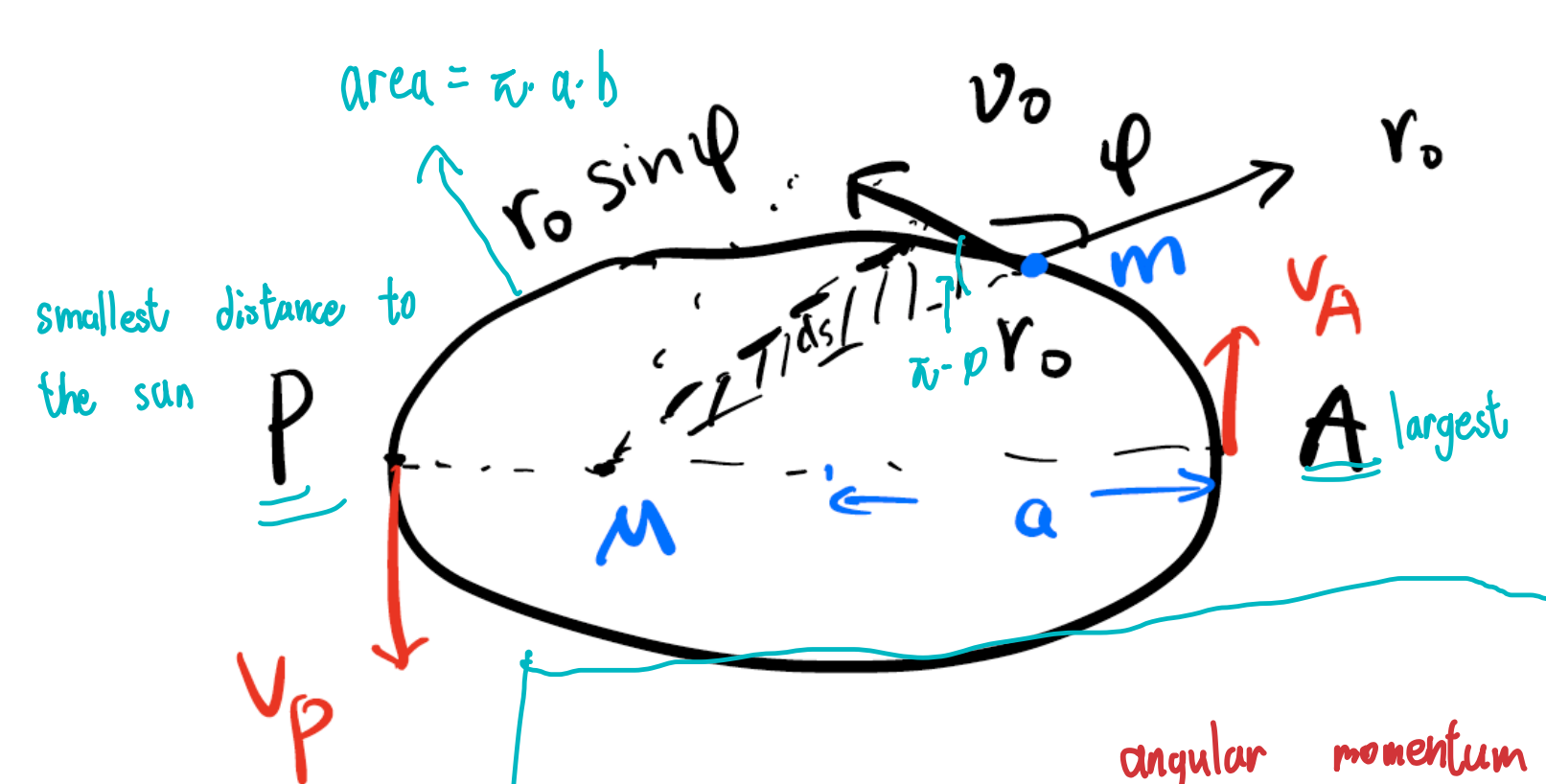
- Third Law: Period proportional to $3/2$ of semi-major axis length

$$T \sim a^{\frac{3}{2}}$$



$$R = a$$
$$\Updownarrow$$
$$T_R = T_a$$

Proof of Kepler's Laws



Known: M, a, m

$T \sim a^{\frac{3}{2}}$
ellipse

$T \sim R^{\frac{3}{2}}$
circular

Find: $v_p, v_A, T, K+U$

Second

angular momentum

$$\vec{\tau} = 0 \Rightarrow \vec{L} \text{ conserved}$$

$$|\vec{L}| = |m\vec{v}| \cdot |\vec{r}| \cdot \sin \phi$$

$$= m v_0 \cdot r_0 \sin \phi = \text{const}$$

$$ds = \frac{1}{2} \cdot v_0 dt \cdot r_0 \sin \phi$$

$$= \frac{1}{2} v_0 r_0 \sin \phi dt$$

$$\frac{ds}{dt} = \frac{1}{2} v_0 r_0 \sin \phi = \frac{|\vec{L}|}{2m} = \text{const.}$$

time same $\Leftrightarrow$ area same
 $\therefore \text{const.}$

$$\frac{ds}{dt} = \frac{1}{2} v_A \cdot (1+e) \cdot a$$

$$= \frac{1}{2} \sqrt{(1-e^2)} \cdot G M a = \text{const.}$$

$$S = \pi a b = \pi a \cdot a \sqrt{1-e^2} = \pi a^2 \sqrt{1-e^2}$$

$$T = \frac{S}{\frac{ds}{dt}} = \frac{\pi a^2 \sqrt{1-e^2}}{\frac{1}{2} \sqrt{(1-e^2)} \cdot G M a} = \frac{2\pi}{\sqrt{GM}} \cdot a^{\frac{3}{2}}$$

$T \sim a^{\frac{3}{2}}$
Conservation of $\vec{L}$ (second law)

$$v_A (1+e) a = v_p (1-e) \cdot a$$

Conservation of E:

$$\begin{cases} E_A = \frac{1}{2} m v_A^2 - G \frac{Mm}{(1+e)a} \\ E_p = \frac{1}{2} m v_p^2 - G \frac{Mm}{(1-e)a} \end{cases}$$

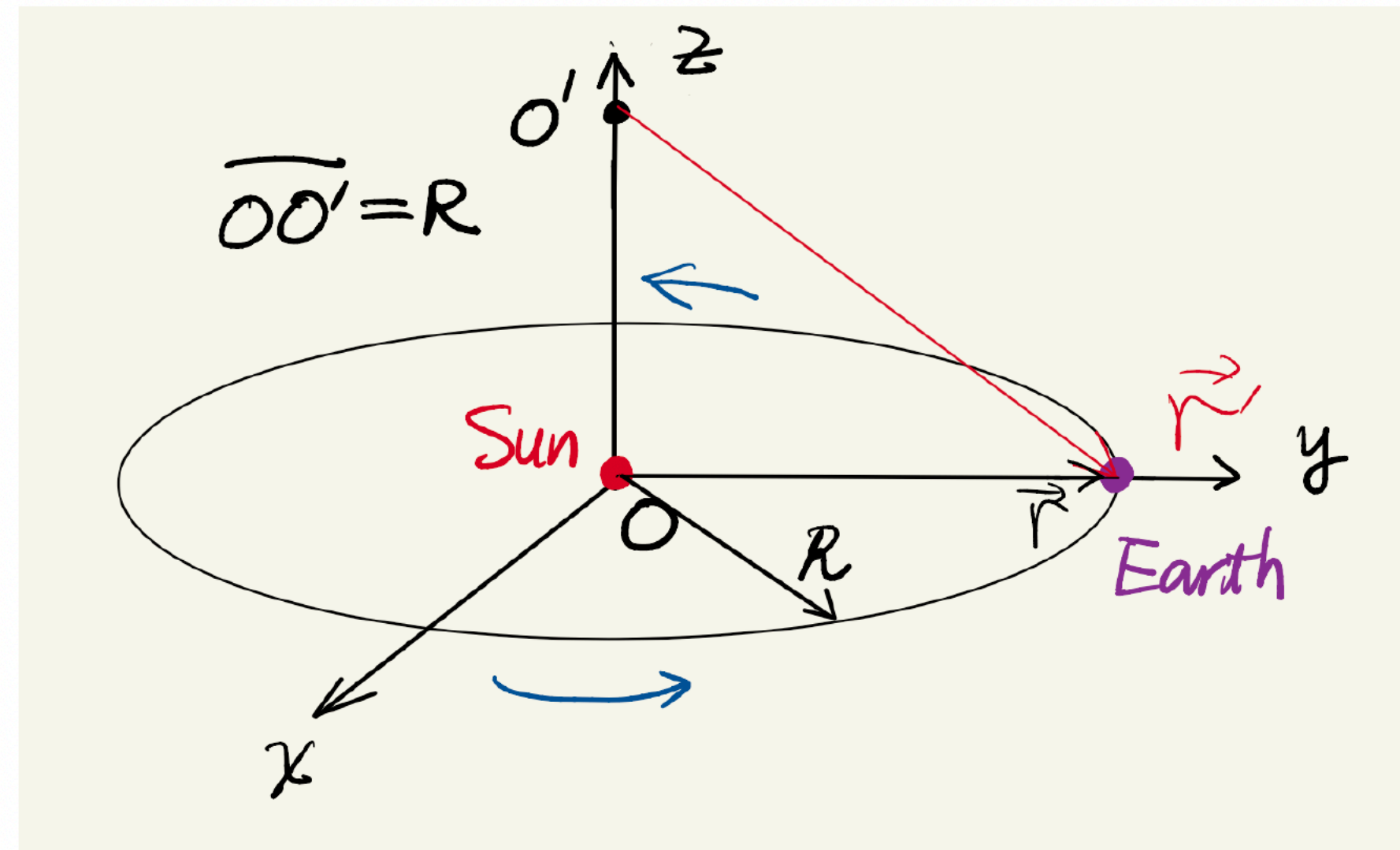
$$v_A = \sqrt{\frac{1+e}{1-e}} \cdot \frac{GM}{a}$$

$$v_p = \sqrt{\frac{1-e}{1+e}} \cdot \frac{GM}{a}$$

Q10, Final 2024

10. Orbital Angular Momentum of the Earth

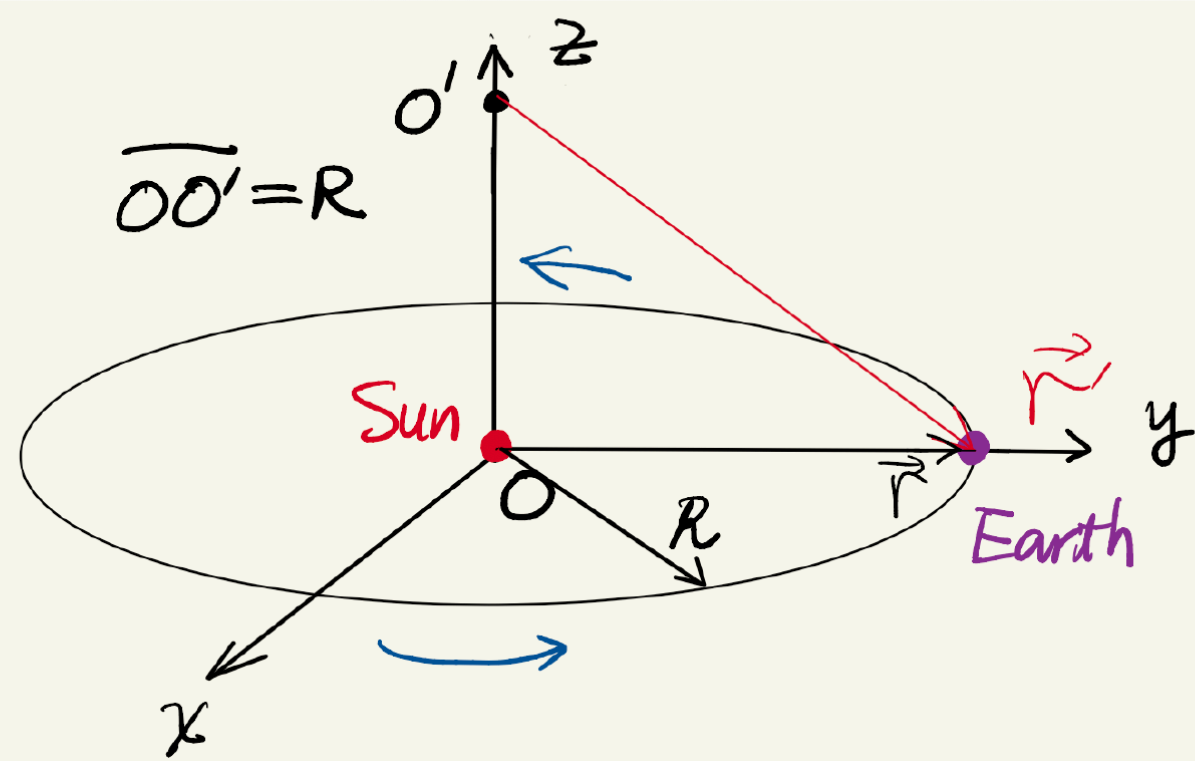
(10 pts)



Consider the Sun-Earth system with masses M_S and M_E , assuming $M_S \gg M_E$ and neglecting other planets and moons. Treat the Earth as a point particle and its orbit as a circular orbit with a radius of R . In the diagram above, the Sun is located at point O , while point O' is directly above O at a distance of R . At one certain moment, the earth is on the y axis, and moving toward the $-x$ direction. (You may refer to this position when you solve for part (c) and part (d)).

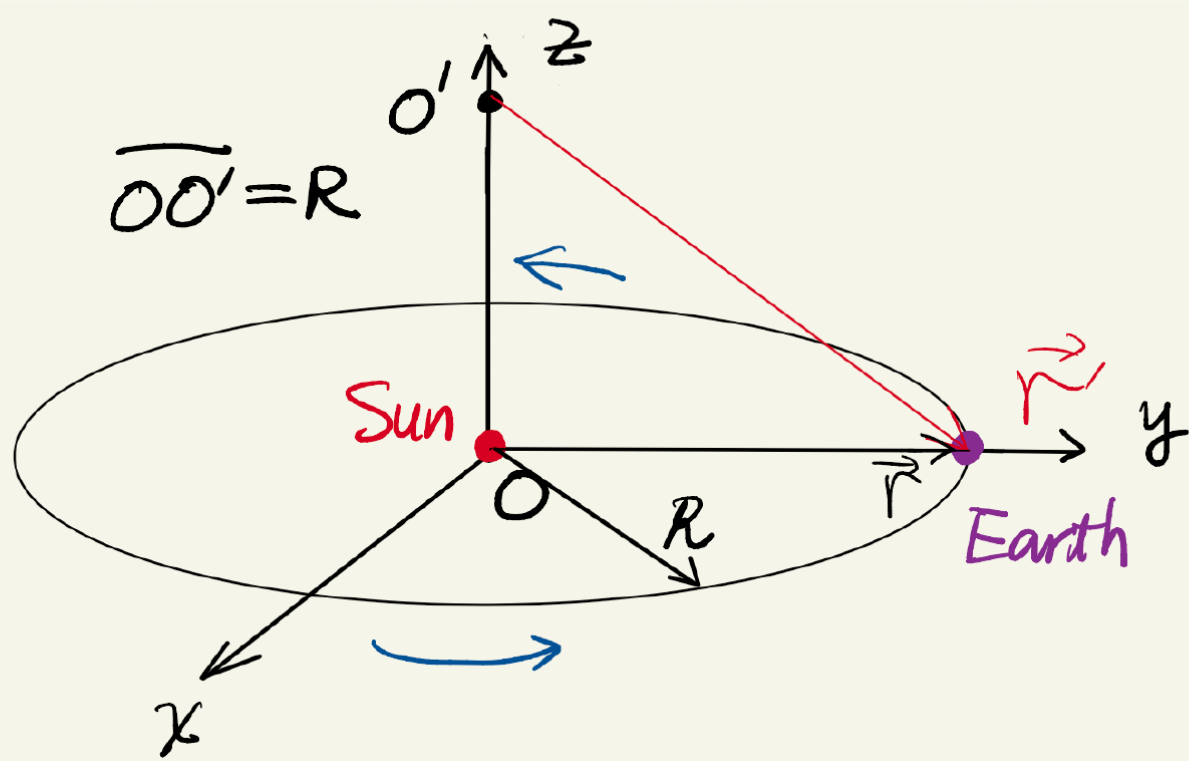
(a) Find the period T_0 it takes for the earth to finish one turn (circle).

(2 pts)

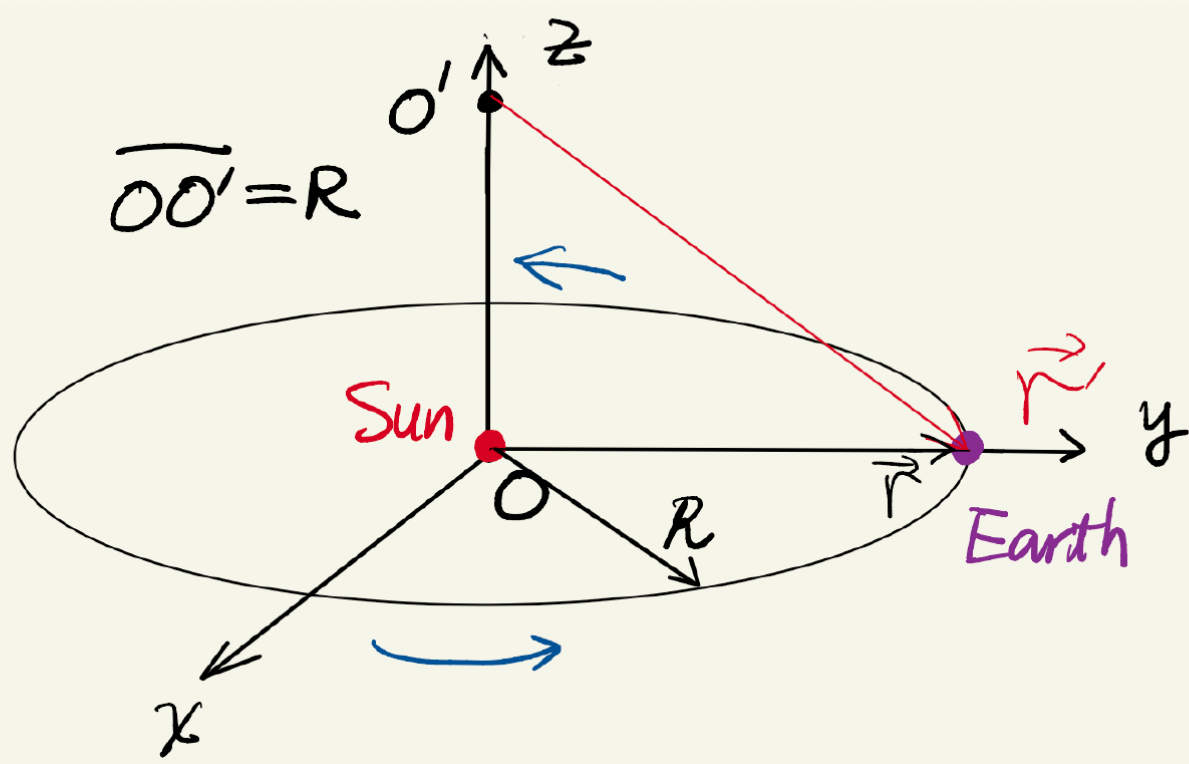


(a) Find the period T_0 it takes for the earth to finish one turn (circle).

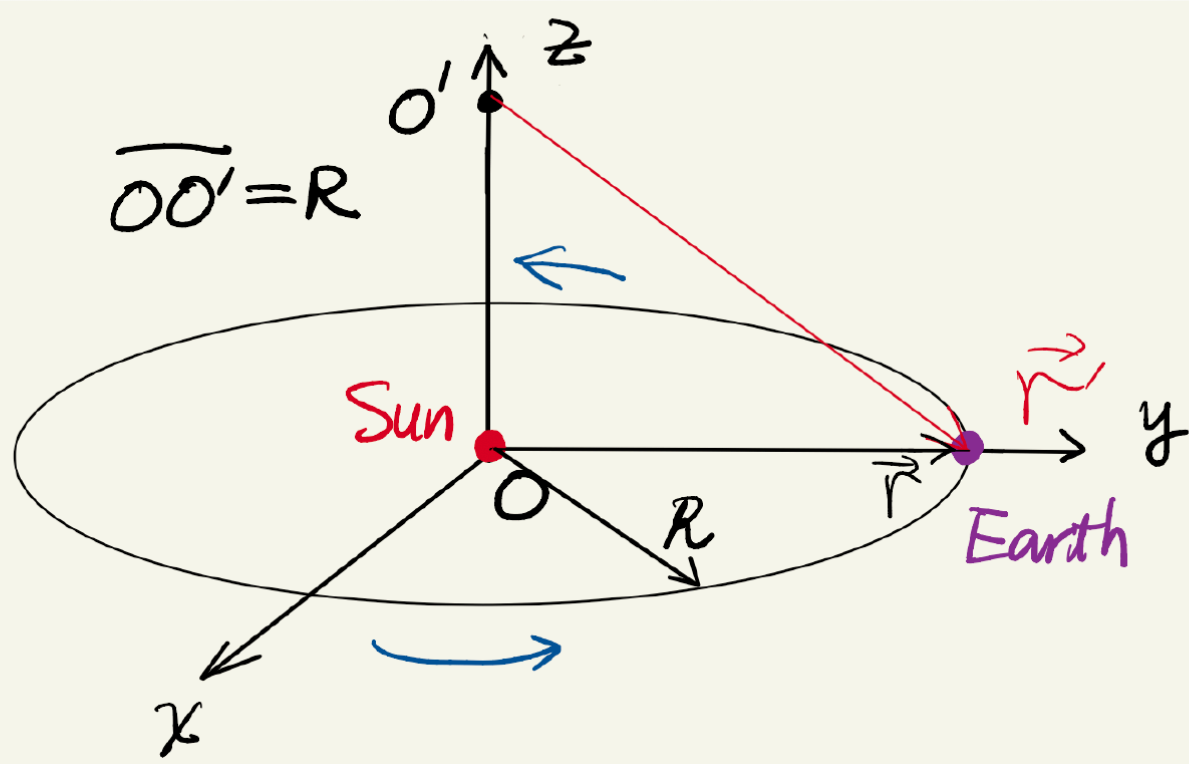
(2 pts)



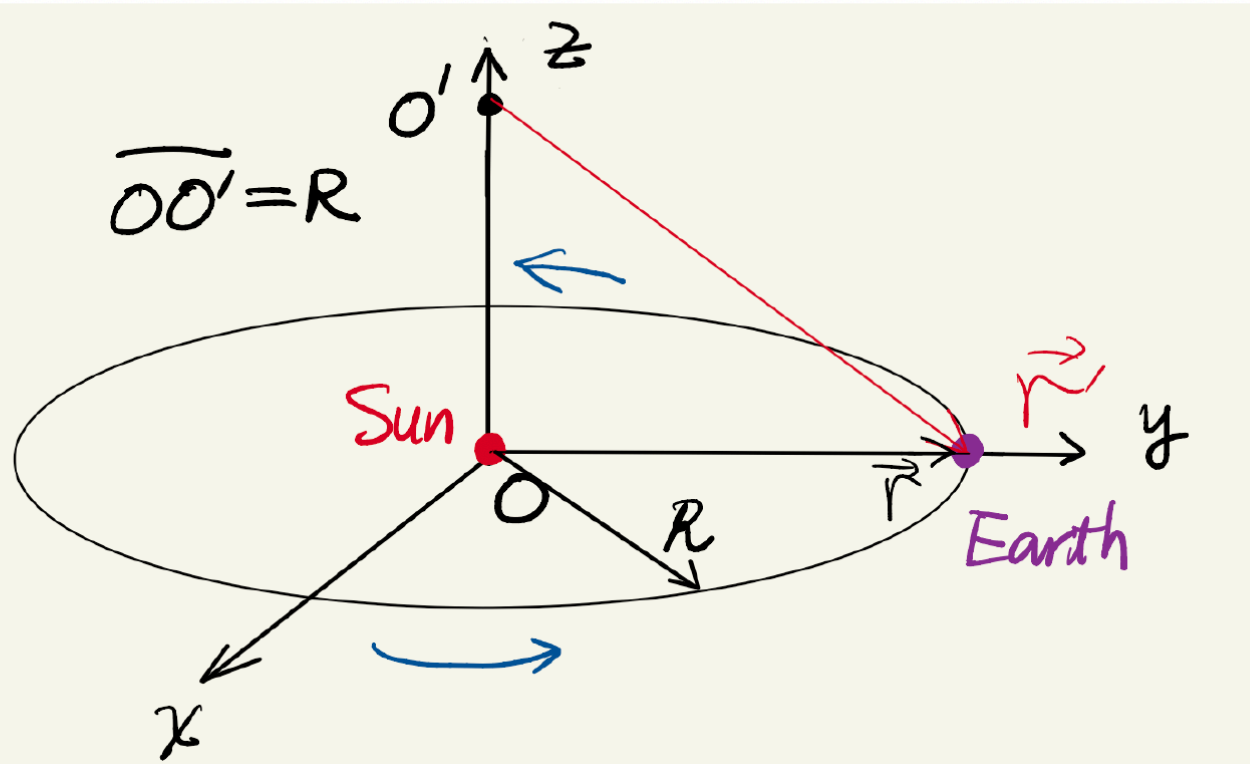
- (b) Choose O as the reference point and compute the orbital angular momentum $\vec{L}_0$ of the earth. Hint: Provide both the magnitude and direction according to $\vec{L}_0 = M_E \vec{r} \times \vec{v}$. **(2 pts)**



(c) Now choose O' as the reference point, and compute the orbital angular momentum $\vec{L}$ of the earth. Hint: $\vec{L}$ is different from $\vec{L}_0$. Give the magnitude and draw the direction. **(2 pts)**



- (d) Compute the torque $\vec{\tau} \equiv \vec{r}' \times \vec{F}$ about the reference point O' generated by the gravitational pull (force) of the Sun on the Earth ($\vec{F}$). **(2 pts)**
Hint: Give the magnitude and direction!



- (e) Choose O' as the reference point, then use Newton's second law for rotation $\vec{\tau} = \frac{d\vec{L}}{dt}$ to explain the change of the orbital angular momentum L . Compute the angular frequency Ω of this "rotation" of the orbital angular momentum. **(2 pts)**

Q12, Final 2024

12. The Restricted Three-Body Problem:

(10 pts)

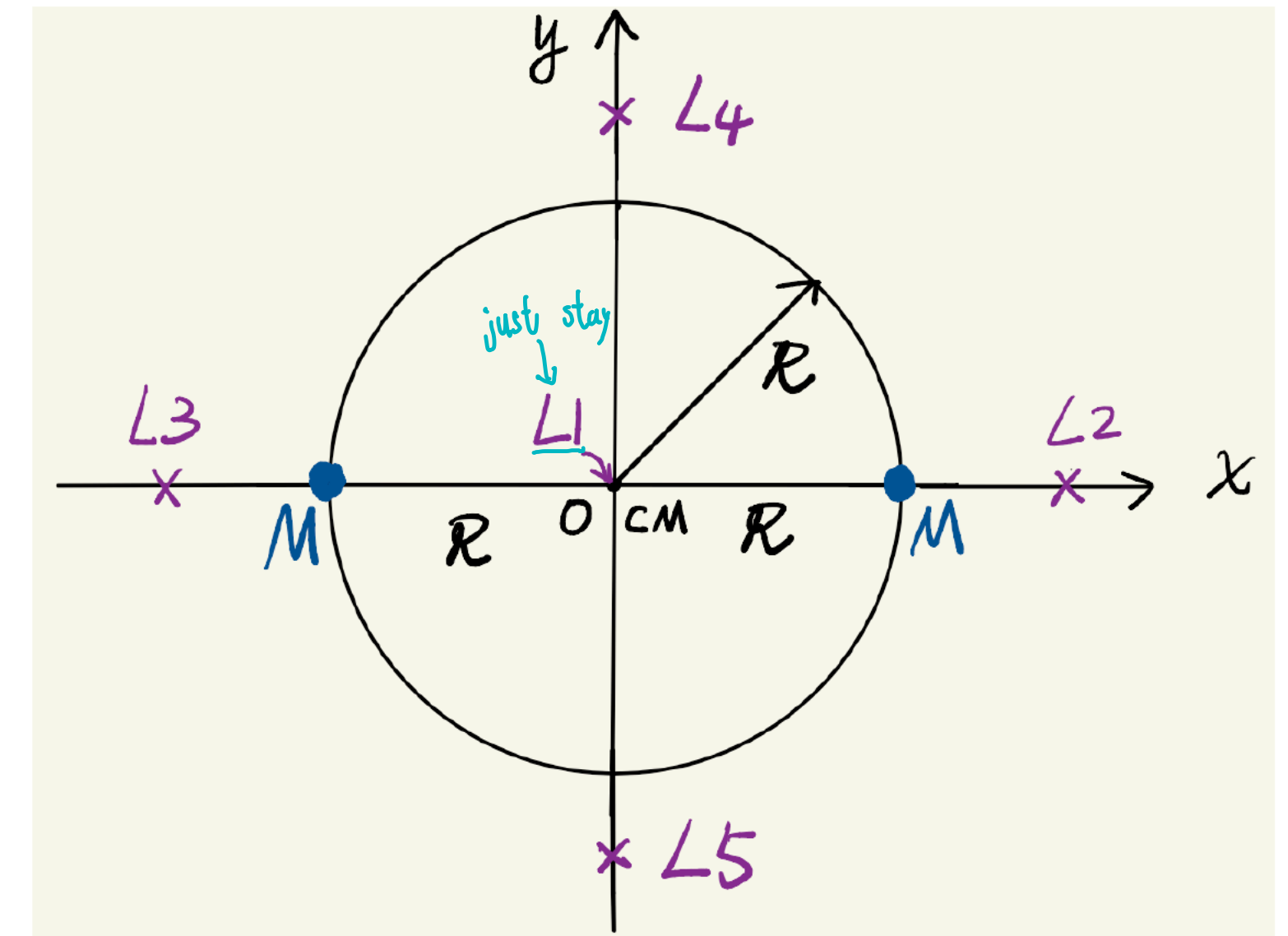
In a galaxy far, far away, the legendary Starship Enterprise discovered a new two-star system: Two equal-mass stars, each with mass M , are rotating about their common center of mass in a circular orbit with radius R .

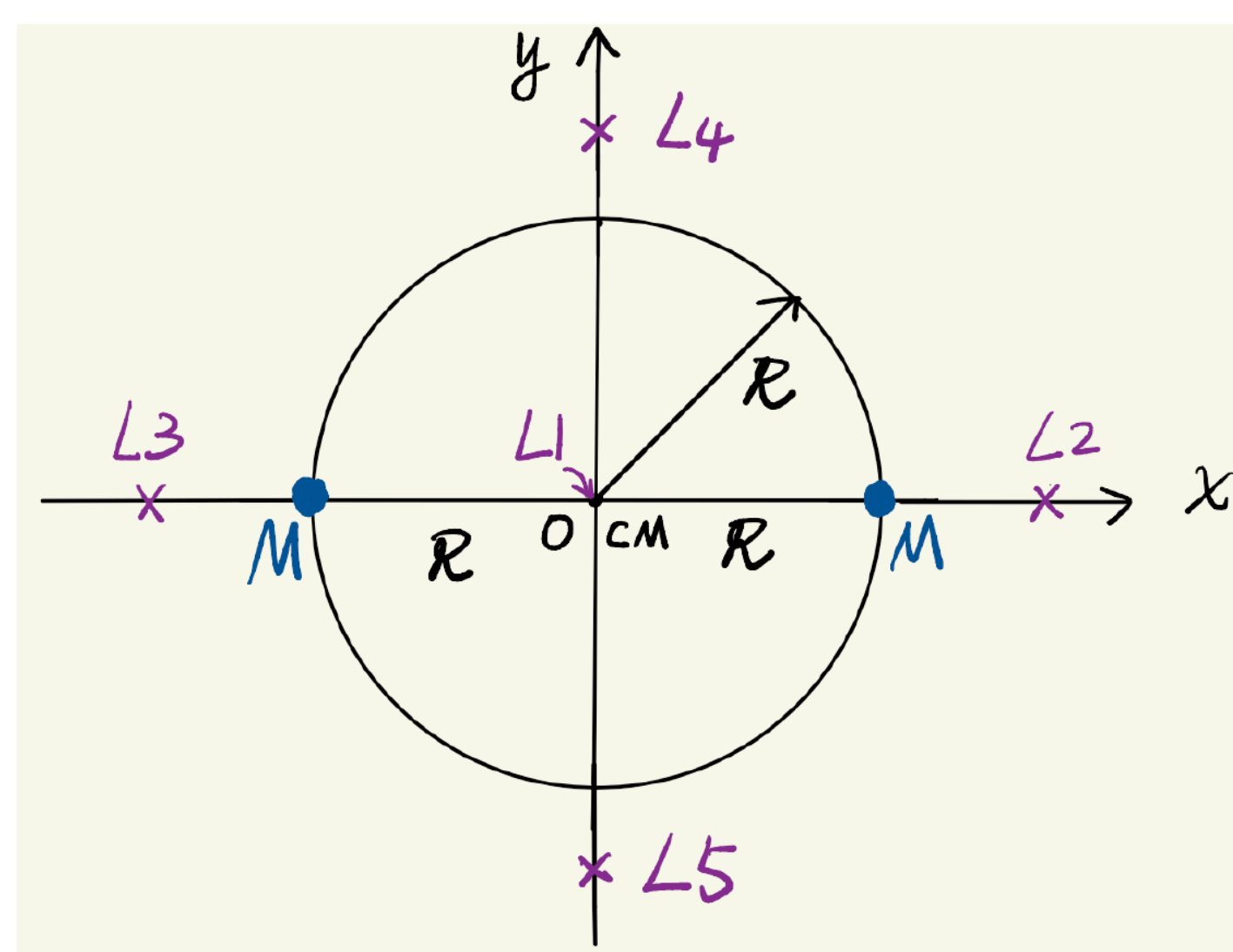
Captain Kirk, renowned for his decisive actions, commands Enterprise to park at a Lagrange point of the two-star system with minimal fuel consumption. At the Lagrange points, the gravitational forces of the two large bodies provide the centripetal force, allowing the Starship to orbit with the same angular velocity as the stars. These three bodies remain in a co-moving configuration, and the Enterprise does not have to use its power.

Your task is to determine the locations of the Lagrange points where Enterprise can be parked under the following three assumptions

- Neglect the gravitational effects of the Starship with mass m on the stars since $M \gg m$.
- Consider the system to be in a two-dimensional plane.
- Assume the stars and the Starship move in circular orbits.

There are five Lagrange points in total. In this special case with two equal mass stars, the first point L_1 locates at the center of mass of these two stars. In the following, we can find the rest of the Lagrange points $L_2 - L_5$.





(a) Find the magnitude of the angular velocity of the two-star system.

(2 pts)

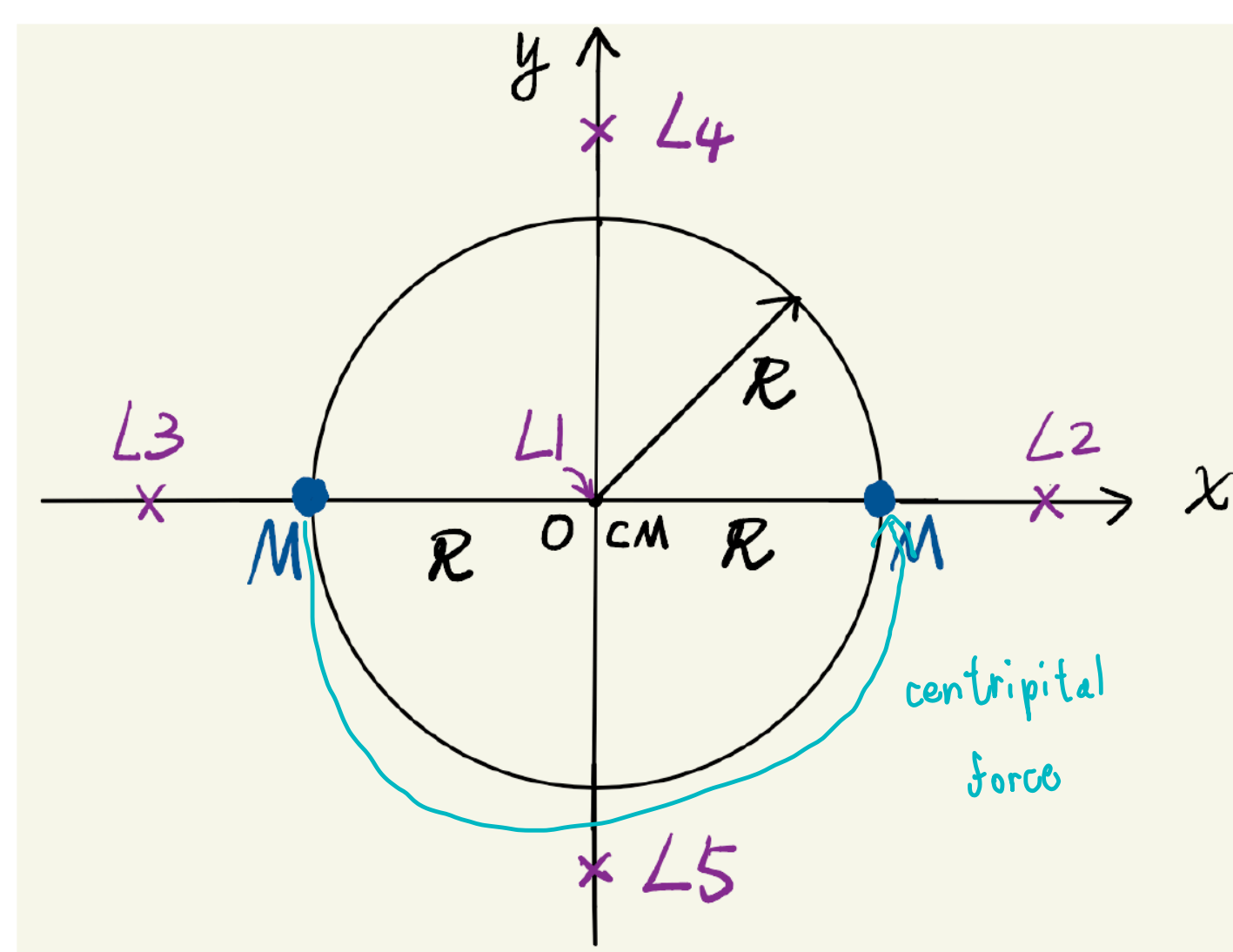
Hint: these two stars are circulating about their CM under the gravitational force $GM^2/(2R)^2$.

$$F_g = G \frac{M M}{(2R)^2} = M \cdot \omega^2 \cdot R$$

$$\omega = \sqrt{\frac{GM}{4R^3}}$$

Two-star system

same ω

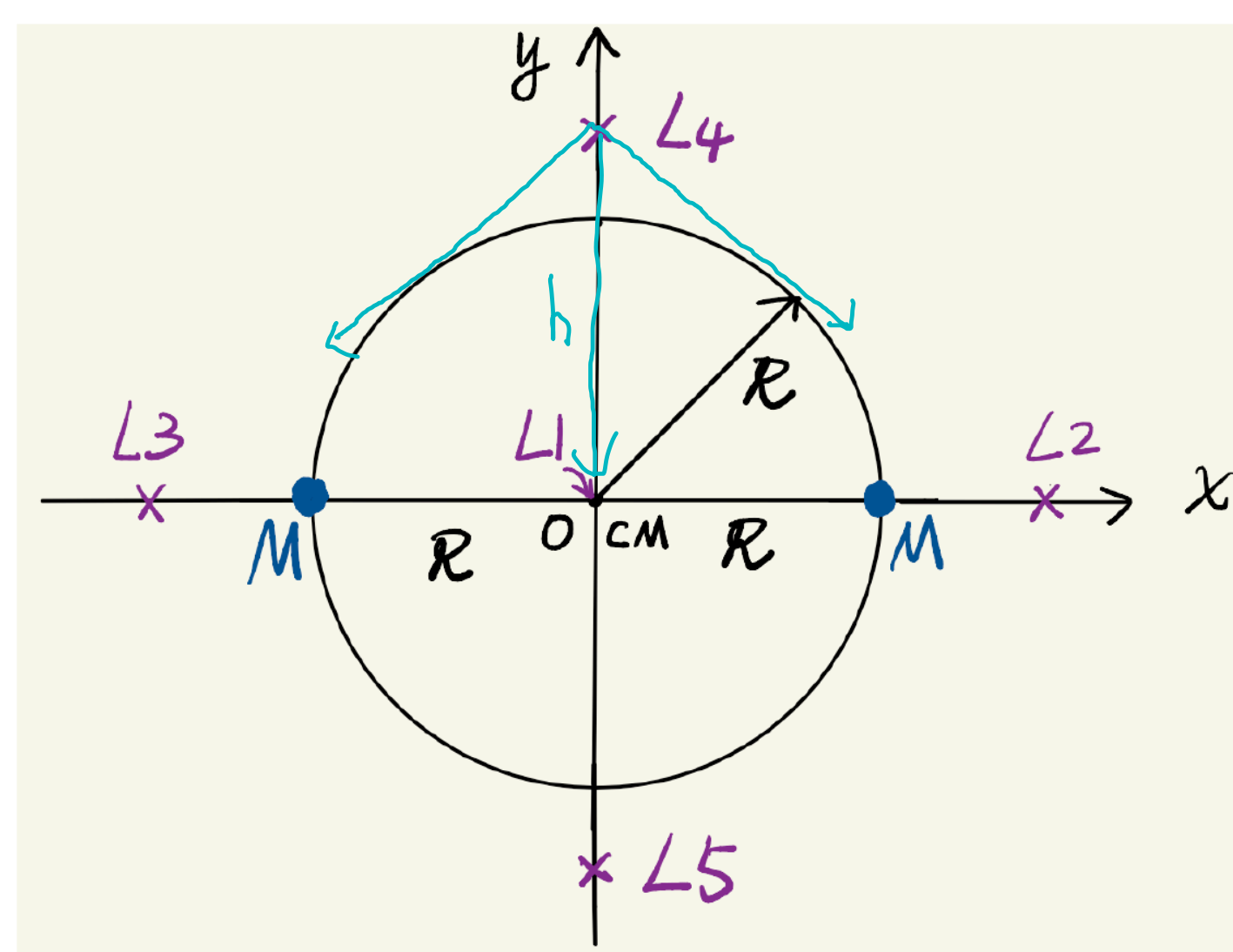


- (b) Find the equation that determines the locations of L_2 and L_3 , which lie on the line through the two stars. **(3 pts)**

(Don't Panic! You do not have to solve this equation during the exam.)

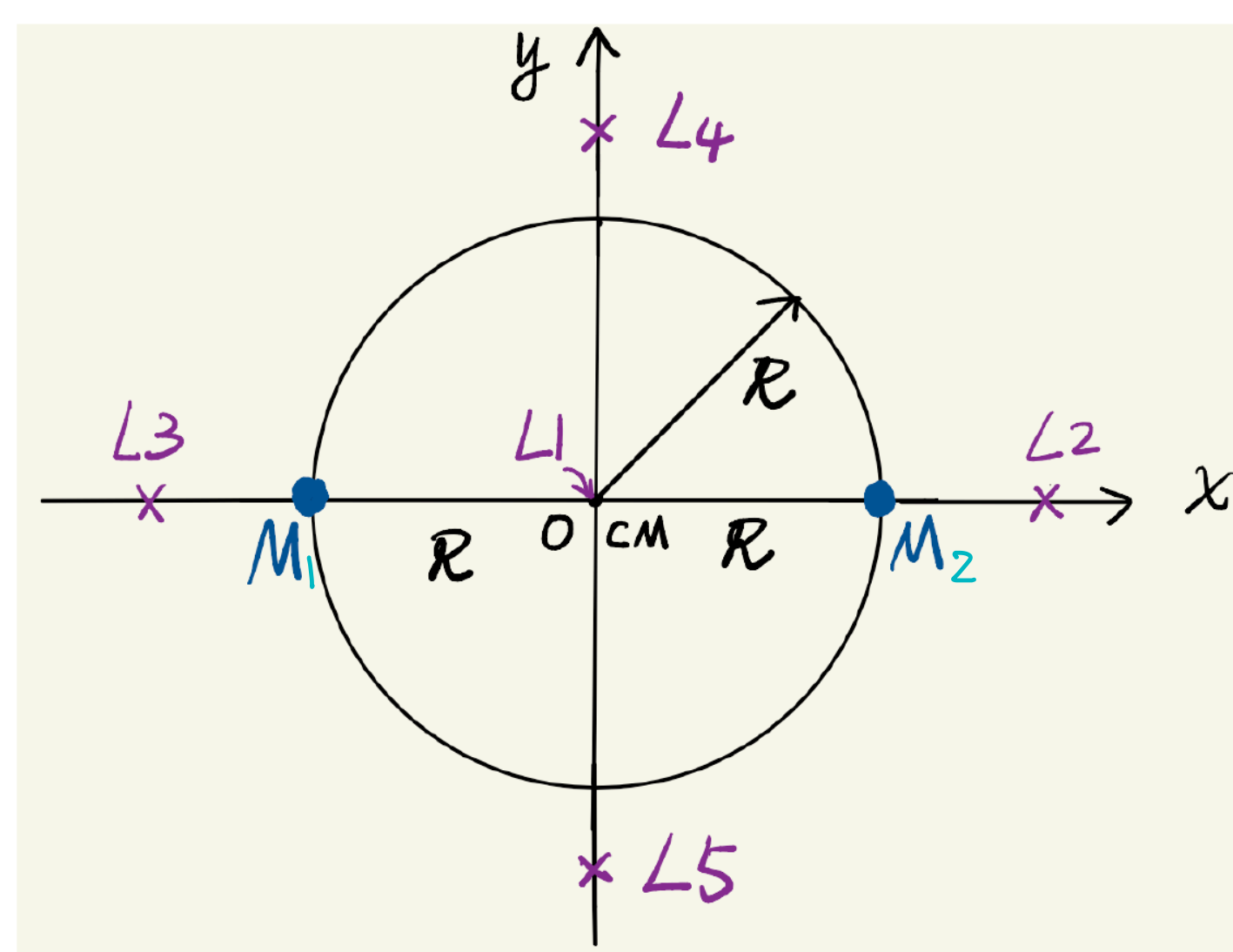
d is negative
 $L_3 \rightarrow L_2 \leftarrow$ change d to positive

$$\frac{1}{(d+R)^2} + \frac{1}{(d-R)^2} = \frac{|d|}{4R^3}$$

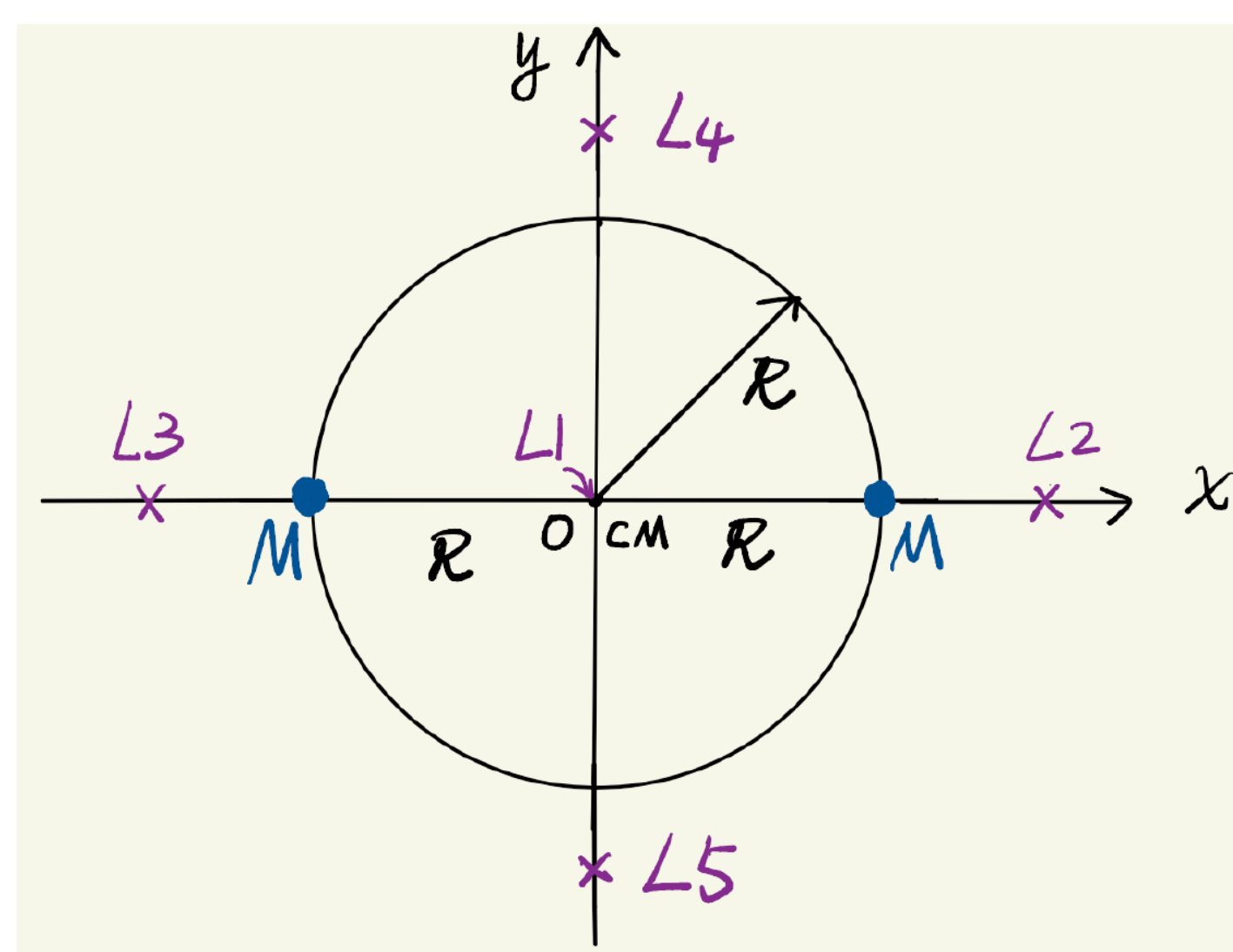


- (c) Find the locations (the y coordinates) of the L_4 and L_5 , which are on the line perpendicular to the line through the two stars. **(3 pts).**

$$F = ma$$



- (d) Now consider the case that these two stars have different masses such as M_1 and M_2 and assume the distance between these two stars is still $2R$. Find the magnitude of the angular velocity again by repeating the calculation in Part (a). **(1 pt)**



(e) Following part (d), will the locations of L_4 and L_5 change as compared to the result in part (c)? (Yes/No) **(1 pt)**

Hint: You do not have to give a reason. A Yes/No answer is sufficient.